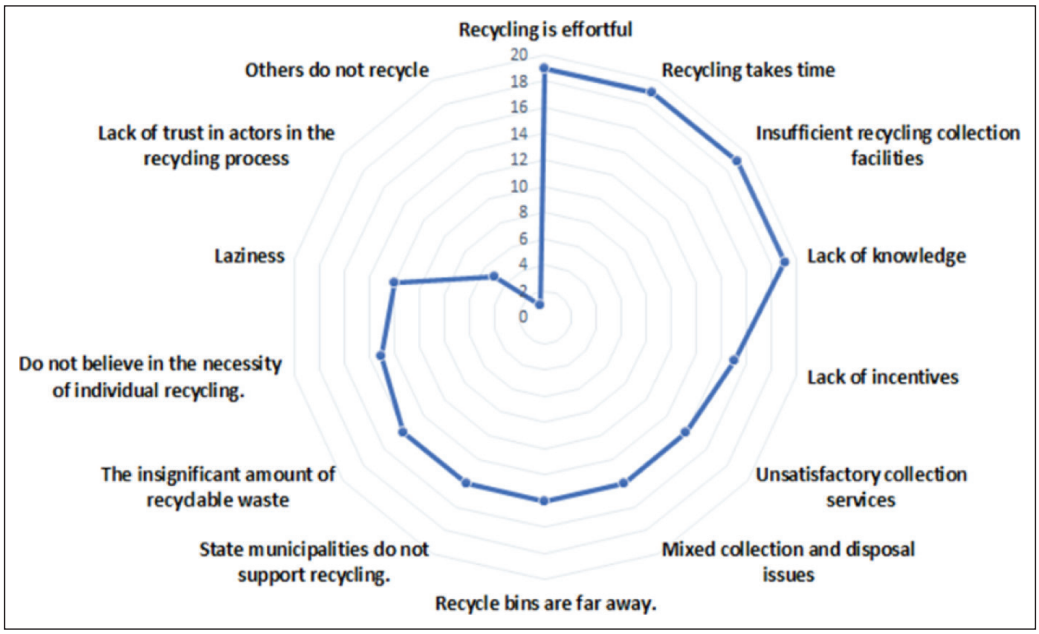


Figure 2. Barriers that can be removed or alleviated by recycling apps.

has a positive and significant correlation with recycling behavior. From the opposite perspective, "lack of knowledge" on what, why, and how to recycle is among the most common barriers to individual recycling in the literature [e.g., 28–30]. Information provided by the apps helps to remove the "lack of knowledge" barrier since users get a deeper understanding of the purpose and positive outputs of individual recycling, identify recyclable materials clearly, and learn how to recycle appropriately. App-provided information is more likely to promote behaviors than conventional communication campaigns since app users are a specific target group already interested in and intent on recycling. However, there is room for functional development for apps through tailoring the information according to the needs of the user. As stated by [31] information tailored according to the characteristics and needs of a specific target group will be more effective in promoting the behavior. However, apps provide generic information for all users. To increase the effectiveness of the recycling apps, they can customize the information provided to users after discovering their primary intentions, beliefs, and needs for recycling. Besides, new ways of information giving can be employed. For instance, an interactive training module customized for specific user segments can be more effective for recycling education. Short video content can also be beneficial in demonstrating visually how recyclable waste should be separated and stored. Furthermore, if videos feature reliable celebrities or opinion leaders, videos not only give information about recycling but also inspiration for recycling.

Some recycling apps also regularly inform users about municipalities' investments, activities, events, waste management, and recycling achievements. In this way, apps may help overcome the barrier of distrust in the recycling actors about the proper collection, care of, and management of recyclables [24, 32]. The removal of this barrier is crucial since the feeling of distrust in recycling actors makes recycling a meaningless action for consumers.

Finally, two applications were found to enable reverse information flow from users to app owners through the reporting function. The function allows users to inform the authorities about the problems in recycling facilities. Although the reporting function is not directly related to the behavior, it may still provide distal support for recycling because contributing to the solution of the problem may increase users' involvement in recycling. More importantly, a quick solution to the reported problem can strengthen participants' trust in the system and alleviate the "lack of trust in actors in the recycling process."

The Role of Apps in Providing Feedback

Providing feedback about individual recycling performance is another function of recycling apps. Users are given statistics about the amount of material they collect for a period. Thanks to this function, users can see that recyclable wastes, even low in quantity, create significant waste when collected after a certain period. So, the function may alleviate the barrier of "the insignificant amount of recyclable waste" [28, 33, 34]. This barrier appears when consumers argue that they produce a minimal amount of recyclables and yields to the belief that recycling is unnecessary. More importantly, feedback about users' recycling performance may promote their perceived behavioral control, which refers to the conviction that they have the ability to recycle their waste. The Theory of Planned Behavior [35] indicates that perceived behavioral control increases recycling both directly and incorrectly. Furthermore, by adding a new function to the apps, users can be given the opportunity to set daily, weekly, or monthly recycling goals; notifications can be sent to encourage users to achieve these goals and give feedback about their final achievement.